

ALI KERIM SECENER

Date of Birth - 02/03/1994

Nationalities - French, Turkish

🏠 Max Delbrück Center for Molecular Medicine (MDC)

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PROFILE

Chemistry and biotechnology engineer with a Ph.D. in Systems Biology and a strong interest in data-driven biology. I work as a bioinformatician, with a background in molecular biology and a focus on single-cell and spatial omics data analysis, including RNA-seq-based technologies and perturbation experiments. I enjoy working at the interface of computation and biology and turning complex datasets into clear, biologically meaningful insights.

WORK EXPERIENCE

- **PhD | Berlin Institute for Molecular Systems Biology (BIMSB), Max Delbrück Center for Molecular Medicine – Berlin, Germany**
JANUARY 2019 - FEBRUARY 2026
TRANSCRIPTOMIC INSIGHTS INTO MOUSE ADRENAL GLAND DEVELOPMENT AND HUMAN ADRENOCORTICAL TUMORIGENESIS
- **Research Intern | Columbia University Medical Center - New York City, United States**
MARCH 2018 - SEPTEMBER 2018
- **Research Intern | Helmholtz Center for Infection Research - Würzburg, Germany**
MAY 2017 - AUGUST 2017
- **Research Intern | Massachusetts General Hospital - Boston, United States**
JULY 2016 - AUGUST 2016
- **Research Intern | Oxitec LTD - Milton Park, United Kingdom**
JUNE 2015 - AUGUST 2015

PUBLICATIONS

- S. Heider, C. Fischer, **Ali Kerim Secener**, P. Vallecillo-García, G. Kotsaris, Z.G. Meisen, V. Pawolski, C. Giesecke-Thiel, T. Conrad, T. J. Schulz, S. Sauer, S. Stricker. 2026. **The dorsal aortic compartment is a developmental source of brown adipose tissue in mice.** *Nature Communications* doi: <https://doi.org/10.1038/s41467-025-68147-9>
- O. Nonn, O. Debnath, D.S. Valdes, K. Sallinger, **Ali Kerim Secener**, C. Fischer, S. Tiesmeyer, J. Nimo, T. Kuenzer, J. Ulrich, T. Maxian, M. Knöfler, P. Karau, H. Bartolomaeus, T. Kroneis, A. Frolova, L. Neuper, N. Haase, A. Malt, N. Müller-Böttcher, K. Kräker, S. Kedziora, D. Forstner, R. Eils, R. Schmidt-Ullrich, S. Heider, S. Verlohren, C. Stern, M. Sugulle, S. Jones, B. Thilaganathan, T.J. Kaitu'u-Lino, S. Tong, B. Huppertz, A. El-Heliebi, A.C. Staff, F. Coscia, D.N. Müller, R. Dechend, M. Gauster, N. Ishaque, F. Herse. 2022. **Senescent Syncytiotrophoblast Secretion During Early Onset Preeclampsia.** *Hypertension* doi: <https://doi.org/10.1161/HYPERTENSIONAHA.124.23362>
- B. Altieri*, **Ali Kerim Secener***, S. Sai, C. Fischer, S. Sbiera, P. Arampatzi, S. Kircher, S. Herterich, L.S. Landwehr, S.N. Vitcetz, C. Braeuning, M. Fassnacht, C.L. Ronchi, S. Sauer. 2024. **Single-nucleus and spatial transcriptome reveal adrenal homeostasis in normal and tumoural adrenal glands.** *Clinical and Translational Medicine* doi: <https://doi.org/10.1002/ctm2.1798>
- D.S. Tourigny, B. Altieri, Ali Kerim Secener, S. Sbiera, M.P. Schauer, P. Arampatzi, S. Herterich, S. Sauer, M. Fassnacht, C.L. Ronchi. 2024. **Cellular landscape of adrenocortical carcinoma at single-nuclei resolution.** *Molecular and Cellular Endocrinology* doi: <https://doi.org/10.1016/j.mce.2024.112272>
- HA. Dinh, M. Volkert, **Ali Kerim Secener**, U.I. Scholl, G. Stölting. 2024. **T- and L-Type Calcium Channels Maintain Calcium Oscillations in the Murine Zona Glomerulosa.** *Hypertension* doi: <https://doi.org/10.1161/HYPERTENSIONAHA.123.21798>
- K. Masuda, A. Kornberg, J. Miller, S. Lin, N. Suek, T. Botella, **Ali Kerim Secener**, AM. Bacarella, L. Cheng, M. Ingham, V. Rosario, AM. Al-Mazrou, SA. Lee-Kong, RP. Kiran, KS. Yan, M. Stoeckius, P. Smibert, A. Del Portillo, PE. Oberstein, PA. Sims, KS. Yan, A. Han. 2020. **Multiplexed single-cell analysis reveals prognostic and nonprognostic T cell types in human colorectal cancer.** *JCL Insight* doi: <https://doi.org/10.1172/jci.insight.154646>



EDUCATION

PhD in Systems Biology | Max Delbrück Center for Molecular Medicine, Berlin Institute for Medical Systems Biology, Genomics Technology Platform

PhD thesis under the supervision of Dr. Thomas Conrad, Dr. Sascha Sauer, and Dr. Altuna Akalin

Biotechnology and Chemistry engineering | Ecole Superieure de Biotechnologie Strasbourg (ESBS) and Ecole Européenne de Chimie, Polymères et Matériaux (ECPM)

Engineering degree linking chemistry and biotechnology fields with provided classes like immunology, analytic, bio-organic and inorganic chemistry, genetic engineering, structure and computational based drug design.

Biology engineering | Ecole Polytechnique Universitaire de Lille

Pre-engineering degree providing preparatory classes like mathematics, thermochemistry, ecology, cell biology and organic chemistry



CONFERENCE PRESENTATIONS

“A multi-modal investigation into the human adrenocortical tumors” Single Cell Biology - Virtually held by the Wellcome Genome Campus; November 2020



Skills

- Wet lab skills: qPCR, cell culture, single cell/spatial technologies (10X Genomics, Cartana), NGS, FACS
- Dry lab skills: R/Python. Proficient in secondary single-cell/spatial RNA seq analysis



Languages

- French: native
- Turkish: native
- English: Full professional proficiency, TOEIC score: 980/990 - 15-Nov-2017
- Spanish: Limited working proficiency
- German: Elementary working proficiency



Achievements & Awards

- Awarded the **55th Schoeller-Junkmann (2022) price**, for the manuscript entitled “Cell Atlas at Single-Nuclei Resolution of the Adult Human Adrenal gland and Adrenocortical Adenomas”